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What Unlinkability Costs at Agent Action Rates: A Composed-Path Measurement and the Soundness of Pre-Computation

The obvious construction misses the budget, and the fix is a security question nobody has asked.

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ABSTRACT

A desk study assessed two numbers against a 15.00 ms per-action budget for anonymous agent credentials — BBS+ presentation at about 13.80 ms, verification at about 4.20 ms — and called each acceptable, because each is below 15.00 ms on its own. Summed, they are 18.00 ms, which is over. With cached pairing pre-computation the sum is 14.90 ms: under the limit by 100 µs, a margin this paper's tool reports as `MARGINAL` rather than a pass, because 100 µs is not headroom for the application, the network, or writing the evidence record. We built `occultation`, a Rust tool that assesses presentation and verification as one composed path rather than two separate ones, over real BBS+ signatures on BLS12-381 and a real Ed25519 baseline. On our hardware the absolute numbers are roughly an order of magnitude smaller than the desk study's, and — the finding the magnitude alone would hide — the *shape* of the correction survives the hardware change while the shape of an unrelated optimization does not: cached pairings save 74 % of verification in the desk study's figures and 11 % of it in ours. The composition error is arithmetic, not a property of any one machine. Pre-computation is what makes the desk study's own published figures fit the budget at all, which converts a performance question into a security one: what happens when the pre-computed pool runs out under burst? We simulate it. The pool stalls; it does not reuse, and reuse is not configurable, because we show precisely what reuse costs — an observer of two presentations sharing one blinding factor recovers the signature scalar and every undisclosed attribute by subtraction, which our test suite performs rather than merely asserts. Stalling is not free either: under a sustained 5000 s^{-1} burst against a ∼1429 s^{-1} refill rate, the worst response in a 2.0 s window is 82 591× an idle one, a timing side channel with a duration-invariant signature — overload factor 3.50×, growth per request 8.33× an idle response, opening at request 89. Separately, we partition a fleet description — a synthetic, hand-authored fixture, not telemetry from any live fleet — to see how large an anonymity set the hardware attestation layer actually leaves standing: 120 identical hosts, after six weeks of the kind of operational drift in patch level and microcode such a fleet would accumulate, partition into 5 distinguishable groups of 71, 34, 11, 3, 1 — 1 host alone on its configuration before any credential is even presented. Effective set size 52.73 of 120; entropy 1.470 bits. Two of the five layers this paper's subject construction proposes — hardware attestation and threshold escrow — are implemented here only

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as cost-representative stubs with no security whatsoever, and every number that touches them says so in the sentence that states it, not in a footnote.

Artifact. `occultation` [1] is Apache-2.0. Every number in Section 5 is regenerated from the tool by `scripts/regen-results.sh` and read into this document from the generated files under `../results/`; continuous integration regenerates the host-independent half of that directory and fails on a diff, so a stale number cannot outlive a code change. The other half is wall-clock timing, differs by host, and is labelled as such everywhere it appears.

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1 The Problem

The Proof-of-Control standard's identity domain, C5, has six requirements — the thinnest of its six domains — and carries open issue 2 [2]: whether an autonomous agent's actions can be independently verifiable and attributable when it matters, while being unlinkable across relying parties the rest of the time. A working-group member raised it in the plainest possible terms: enterprises need to deploy agents without disclosing customer PII or corporate identity to third-party vendors, and the best way to secure information is not to hand it out unnecessarily.

The tension is structural, not incidental. Every other part of the standard this domain sits inside pushes toward *more* identifiable evidence: an explicit `agent_id`, an `initiating_user`, a hash chain that links every action to every action before it, anchors published where anyone can see them. A hash-chained, publicly anchored log of an agent's actions is close to the worst case for unlinkability: it is a permanent, ordered, cryptographically bound dossier. Making that unlinkable is not a matter of omitting a field from a record. It requires a credential layer whose presentations do not link to each other, sitting underneath a log layer built, by design, to chain everything together.

A desk study assembled a concrete five-layer answer — hardware attestation, an anonymous credential, an evidence log, a threshold escrow for accountability, and a transport layer — and estimated what it would cost, using published benchmarks of other people's hardware rather than measuring anything. The estimate contained an arithmetic error large enough to flip its own headline conclusion, and finding that error is most of what this paper is about. The desk study assessed presentation and verification against the standard's 15.00 ms per-action budget *separately*: presentation at about 13.80 ms, under budget; verification at about 4.20 ms, also under budget. Both halves passing is not the same claim as the sum passing, and the sum — 18.00 ms — does not. No cryptographic assumption in the desk study is wrong. The mistake is in the addition, and it survives being pointed out only because presentation and verification happen on different machines at different times, which makes it easy to grade them on separate report cards.

This paper reports what happens when that composed path, and the two security questions the desk study's own numbers raise but do not answer — whether the fix for the composition error is itself sound, and how large an anonymity set the hardware layer actually leaves standing — are measured rather than estimated.

2 What Others Have Built

The mechanisms this design needs are mature. Applying them to a machine-speed, hash-chained, publicly anchored agent log is not, and every citation in this section is doing double duty: establishing that the primitive exists and is trusted, and marking the gap between that primitive and this application.

Hardware attestation without a platform identifier. Direct Anonymous Attestation [3] lets a TPM prove that *some* valid TPM produced a statement without revealing which one. It is adopted in TPM 2.0 and standardized as ISO/IEC 20008 [4]. The elliptic-curve variant, ECDAA, is the practical form used in FIDO and is the closest existing answer to the hardware layer this construction needs. It is also the layer we do not implement: Section 3 explains why it is modelled rather than built.

Anonymous credentials. Camenisch–Lysyanskaya credentials [5], Idemix [6], and U-Prove [7] are the lineage this paper's credential layer descends from. BBS+ signatures over pairing-friendly curves [8, 9], with a maturing IETF/W3C track for selective disclosure [10], are the concrete choice the desk study made and the one we implement over BLS12-381. Also relevant, and less well known, is SPSEQ-UC [11], structure-preserving signatures on equivalence classes over updatable commitments: it targets re-randomization directly, which is the property a log layer built from these credentials would need and which our evidence-log layer does not yet touch.

Group and ring signatures. Chaum and van Heyst's group signatures [12] and Rivest, Shamir and Tauman's ring signatures [13] are the other route to "some member of a group signed this." Group signatures include an opening authority, which is the shape our escrow layer generalizes to a quorum.

Unlinkable, bounded tokens. Privacy Pass [14] and the VOPRF construction beneath it [15] are the closest deployed analogue to "unlinkable but rate-limited," and the desk study's own cost table lists a Privacy Pass token at roughly 850 µs, sub-ms amortized when pre-fetched in batches — cheap enough that it is plausibly the mechanism that makes an unlinkable design practical rather than merely possible, though it is out of scope for the construction this paper measures.

Threshold cryptography and escrow. Threshold ElGamal, distributed key generation, and Chaum–Pedersen DLEQ proofs for verifiable partial decryption [16] are the machinery that turns "escrow" from a standing backdoor into a quorum action with a public receipt. The literature that keeps that machinery honest is the key-escrow critique in *Keys Under Doormats* [17] and the specific, documented failure modes of the Clipper Chip [18]: a centralized depository, fragile operational control, and no transparency. Section 3 returns to this literature when it explains why our escrow stub's absence from the real half of this tool is not an oversight.

Transport-layer linkability. Oblivious HTTP [19] and relay-based padding schemes are easy to forget and fatal to omit: an unlinkable credential presented over a directly attributable TCP connection is not unlinkable. This construction, and this paper, do not touch the transport layer at all; Section 7 says so explicitly rather than leaving it as a silent gap.

Where this paper sits. None of the above measures what BBS+ costs at agent action rates, composed as a single present-and-verify path, against a fixed real-time budget, on real hardware. That measurement — and the arithmetic error it corrects — is this paper's contribution. The security question the correction's own fix raises, whether a pre-computation pool is sound under burst, appears nowhere in this literature either, because in most of it pre-computation is a performance detail rather than the thing standing between the design and its own latency budget.

3 The Construction We Measured

The desk study's construction has five layers.

Not all five layers are implemented, and none of the three that are touched here are implemented to the same standard. Table 2 states the boundary exactly, and it is required to match `occultation`'s `README.md` member for member: a paper's own claim about what it measures or models should be checked against the tool's documentation directly rather than assumed from the

Table 1: The five-layer construction P05 proposes. `occultation` measures or models rows 1, 2 and 4 (hardware attestation, agent credential, accountability); the evidence-log and transport rows are out of scope for this paper (Section 7).

Layer	Mechanism	Defeats
Hardware attestation	ECDAA: prove membership in a valid TEE group without emitting a platform identifier	a relying party or auditor fingerprinting the silicon
Agent credential	BBS+ / SPSEQ-UC: fresh blinding per presentation, selective attribute disclosure	colluding relying parties correlating sessions
Evidence log	ephemeral per-session chains, roots posted as re-randomizable Pedersen commitments	a log operator reconstructing a global action sequence
Accountability	k-of-n threshold ElGamal escrow tag with a NIZK that the tag encrypts the credential-bound identity	a rogue escrow node unmasking unilaterally
Transport	OHTTP relay + padding	a network observer linking sessions by IP and timing

fact that Table 1 names a stub, because not every gap between a claim and the code behind it is about a modelled component specifically.

Three of these five rows are what the rest of this paper touches: hardware attestation, the agent credential, and accountability. Of Section 5's figures, only Table 4 — the agent-credential row, not explicitly labelled `MODELLED` — rests on real BLS12-381 pairing arithmetic and a real BBS+ implementation timed on real hardware. The section's other figures are not timings on this hardware at all: Table 3's desk-study columns are the published study's own constants, never run here; Table 5 partitions a fixed TOML fixture, which involves no cryptography to time; and Section 5.2's burst simulation runs on operator-supplied cost inputs, not a live measurement. Every figure that touches ECDAA or threshold escrow — the hardware-attestation and accountability rows — is a cost estimate from a stub with no cryptographic content, and this paper does not use either stub to support any claim about security — only, where unavoidable, about cost, and always labelled.

The escrow stub's absence from the real half of Table 2 is deliberate, not an oversight, for the reason Section 2's escrow literature makes plain: a k-of-n threshold ElGamal tag with a NIZK of correct encryption is exactly the kind of machinery *Keys Under Doormats* [17] and the Clipper Chip's documented failure [18] warn is easy to build and hard to build safely — the danger is in the governance surrounding a working implementation as much as in the cryptography itself, and a stub built only to be cost-representative cannot accidentally be mistaken for a governance guarantee this paper has not evaluated. Building and asserting the security of a real threshold-opening path is future work, not something a cost-measurement tool should attempt as a side effect.

The evidence-log layer in Table 1 is not implemented at all; it is P05's open questions 1 and 2 (Section 7), and nothing in this paper's measurements bears on it. The transport layer in Table 1 is likewise not implemented, but it is a separate, untouched omission rather than one of those two open questions; Section 7 names it on its own.

Table 2: What `occultation` is, stated as a three-way split, matching `README.md` exactly. "Real but not interoperable" is not a hedge: the cryptography is genuine and every property below it is asserted by a test, but the wire encoding has not been checked against another implementation.

Category	Members
Real	The Ed25519 baseline, the BLS12-381 group and pairing arithmetic, the pre-computation pool, the anonymity-set calculator, and the timing harness.
Real but not interoperable	BBS+ key generation, signing, verification, presentation and proof verification, implemented over `blstrs` BLS12-381 against `draft-irtf-cfrg-bbs-signatures` [10]. Forged signatures fail, tampered proofs fail, and undisclosed attributes stay undisclosed — each asserted by test. It is not validated against the draft's test vectors, so its wire encoding will not interoperate with another BBS+ implementation.
Modelled — no security whatsoever	ECDAA (`ModelledEcdaa`) and threshold ElGamal escrow (`ModelledThresholdElGamal`). Both carry the correct interface and a representative cost profile and provide no anonymity, no soundness, no confidentiality and no accountability. Neither executes without an explicit `--allow-modelled` flag; asking for one without it exits non-zero and explains why, rather than silently running a component that provides no security.

4 Measuring the Composed Path

The desk study's error was structural: it judged two numbers against one budget by comparing each number to the budget separately. `occultation`'s `bench --composed` command exists specifically to make that structural mistake unavailable. It composes a presentation cost and a verification cost into a single `Composed` value, sums them, and judges the sum — never a half — against the 15.00 ms budget defined once, in `src/cost.rs`, as the single source of truth `bench` reads (`pool` and `anonymity` answer different questions and do not consult it). A verdict is one of three values, not two: `Fits`, `Misses`, or — the variant that earns its place — `Marginal`, for a composition that is under budget by no more than 5 % of it. Without that third value, 14.90 ms against a 15.00 ms budget reports as a pass, which is exactly the desk study's own conclusion about its cached-pairing figure and exactly the conclusion this tool exists to correct.

`bench --composed` reports two profiles side by side, never merged. The *desk-study* profile reproduces the published figures verbatim, tagged `PUBLISHED`, with their citation — these numbers are constants, not measured on this stack, and they are reported so that the composition arithmetic is checkable by a third party without access to our hardware. The *measured* profile times the same operations, here, with a seeded RNG and a median of many iterations, tagged `measured here`. The two profiles are expected to disagree in magnitude, because they come from different silicon, and Section 5 reports exactly how much and in what shape they disagree.

One row in the measured profile carries a third tag, `ESTIMATED`, and it needs a precise statement of what it is not. Nothing in `occultation` measures a pooled presentation's *online* phase directly — the work left to do once a blinding factor has already been pre-computed and pulled from the pool. The pooled row in the measured composed-path table is derived from a cheaper proxy, two hash-to-scalar operations, standing in for the challenge hash and field-arithmetic responses a real online phase would perform. `occultation pool`, which is the command that actually exercises

the pool under load (Section 6), does not close this gap either: its `--online` cost defaults to this identical estimate when the operator does not supply one. So the estimate and the pool simulation are not independent evidence for each other, and the row's own note says the estimate is almost certainly too low — it implies a 99 % reduction in presentation cost where the desk study's own pooled figure implies 80 %. We report it because omitting the row would leave the measured profile's pooled composition entirely unstated, and a caveated estimate is more honest than a silent gap. It is not a claim that we have measured a pooled presentation, and no number in this paper should be read as one.

5 Results

Every table and every inline figure in this section is generated by `scripts/regen-results.sh` from files under `../results/` and `\input`-ed or substituted as a LaTeX macro; none of it is typed here by hand. Files under `results/deterministic/` are arithmetic over the desk study's published constants, a seeded pool simulation (including `pool-facts.tex`), or a partition of a fixed fleet file, so they reproduce byte-for-byte on any host and continuous integration fails the build on a diff. Files under `results/measured/` are wall-clock timing, differ by host, and are not diffed for that reason; every number drawn from them is marked as measured, either in its own table or, for `composed-facts.tex`, in the prose that cites it.

5.1 The composed path, corrected

Composition	Present	Verify	Total	Verdict
naive (no pool, no cached pairings)	13.80 ms	4.20 ms	18.00 ms	MISSES by 3.00 ms
cached pairings only	13.80 ms	1.10 ms	14.90 ms	MARGINAL — 0.10 ms spare
pre-computation pool + cached pairings	2.80 ms	1.10 ms	3.90 ms	fits, 11.10 ms spare

Table 3: The composed present-and-verify path against the 15.00 ms budget: the desk study's published figures for presentation and verification, assessed together rather than separately as the desk study assessed them. The naive composition sums to 18.00 ms and misses the budget by 3.00 ms even though each half alone appeared to fit. Caching pairings brings the sum to 14.90 ms, 0.10 ms under the limit — not headroom, hence `MARGINAL` rather than a pass. Only the pre-computation pool, combined with cached pairings, fits with room to spare: 3.90 ms total, 11.10 ms of headroom. `results/measured/composed.txt` reports the profile measured on this host, which diverges from these published figures by roughly an order of magnitude without changing the correction, which is an error in arithmetic, not in hardware.

Table 3 is the desk study's own published figures, composed as one path instead of two. The naive composition misses the budget outright. Caching pairings brings it to `MARGINAL` rather than a pass — under the limit, but by a margin smaller than the jitter of an uncontended context switch. Only the pre-computation pool, combined with cached pairings, fits with room, which is the paper's headline correction: pre-computation is not an optimization on top of a design that already works. It is the reason the design fits its own budget at all.

Table 4 measures the individual primitives that compose Table 3's presentation and verification costs, timed here on this host rather than read from a publication: ten primitive operations rather

than three compositions, and without a budget verdict against any of them individually, which is the entire point Table 3 exists to make.

Operation	Cost	P95	×Ed25519	Provenance
Ed25519 sign	0.007 ms	0.007 ms	0.3	measured here (n=200)
Ed25519 verify	0.018 ms	0.019 ms	0.7	measured here (n=200)
Ed25519 sign + verify (baseline)	0.025 ms	–	1.0	measured here (n=200)
G1 scalar multiplication	0.039 ms	0.040 ms	1.6	measured here (n=200)
hash_to_scalar	0.002 ms	0.002 ms	0.1	measured here (n=200)
BBS+ sign (issuer)	0.208 ms	0.219 ms	8.2	measured here (n=200)
BBS+ signature verify (holder)	0.642 ms	0.698 ms	25.3	measured here (n=200)
BBS+ presentation	0.539 ms	0.564 ms	21.2	measured here (n=200)
BBS+ proof verify	0.766 ms	0.793 ms	30.2	measured here (n=200)
BBS+ proof verify, cached pairings	0.684 ms	0.713 ms	27.0	measured here (n=200)

Table 4: Primitive costs measured on this host against an Ed25519 sign-and-verify round trip. ECDAA and threshold escrow are omitted: they are modelled stubs and this table is generated from a plain `occultation bench` run, which never executes a modelled component. See `results/measured/HEADER.txt` for the exact host and toolchain.

The measured composed path (`results/measured/composed.txt`, reported in prose here rather than as a table) is 14× cheaper here than the desk study's published figures imply, and the composed BBS+ path costs 51× an Ed25519 sign-and-verify round trip on this host. That divergence factor is not an apples-to-apples comparison: the desk study's naive presentation figure folds in a 4.6 ms (33 %) allowance for the escrow tag, while the measured figure here excludes escrow entirely; excluding escrow from both sides instead gives a divergence of roughly 10×. The composition error itself is unaffected by any of this: it is an error in arithmetic, not a property of any one machine, and it reproduces identically wherever the desk study's published figures are summed.

What the magnitude alone would hide is that the divergence is not uniform. Cached pairings are the one optimization measured on both sides of the comparison: the desk study's published figures attribute 74 % of verification cost to caching the pairing computation, and the same optimization saves 11 % here. A uniformly faster host would preserve the desk study's ratio between optimized and unoptimized verification; this host does not, so the measured block does not corroborate the desk study's optimization structure, which we report as a finding in its own right rather than as noise to be averaged away.

5.2 Pool exhaustion under burst

Section 6 argues that the pre-computation pool's soundness, not its speed, is the question the correction above actually raises. The numbers behind that argument, from the shipped burst configuration (capacity 64, refill 700 µs, online 60 µs, burst rate 5000 s^{-1}, seed 7 — the same configuration `tests/acceptance.rs`'s acceptance test 2 exercises) are: of 10 000 requests over a 2.0 s window, 89 were served immediately and 9911 stalled. The worst response was 82 591× an idle one. That figure is a peak over the observation window and grows linearly with it, so we report it beside its duration-invariant companions: an overload factor of 3.50× arrivals over refill, a growth rate of 8.33× an idle response added to each successive stall once the pool runs dry, and an onset at request 89 — how briefly a burst has to be sustained before the channel opens. Every one of 10 000 requests received a distinct blinding factor; reuse events: 0.

The refill and online costs above (700 µs and 60 µs) are values `scripts/regen-results.sh`

stipulates for this simulation, not a measurement of any specific deployment; 82 591× and 8.33× both scale inversely with the chosen online cost, while 3.50× does not depend on it.

5.3 Anonymity under fleet drift

Size	Host group	PCE SVN	Microcode	TDX module
71	us-east-pool-patched	13	0x2b000603	1.5.05
34	us-east-pool-pending	12	0x2b000603	1.5.05
11	eu-west-rack-7	12	0x2b000401	1.5.03
3	us-east-canary	13	0x2b000701	1.5.05
1	us-east-rebuilt-0419	14	0x2b000701	1.5.06

Table 5: Anonymity sets for fleet `acme-prod-drifted`: 120 hosts of identical hardware, after six weeks of ordinary operational drift in patch level, microcode and PCE SVN. 5 distinct TCB configurations emerge; the largest anonymity set is 71 of 120 hosts, and 1 host(s) are alone on their configuration — no credential scheme can help them, because the quote collateral identifies the machine before any credential is even presented. Effective set size 52.73 of 120; entropy 1.470 bits.

Table 5 answers a question the desk study's cost analysis does not ask at all: given perfect unlinkability at the credential layer, how large is the anonymity set the hardware attestation layer actually leaves standing? `examples/fleet-drifted.toml` is a synthetic, hand-authored fixture, not telemetry from any live fleet; it encodes 120 hosts of identical hardware after six weeks of the kind of operational drift a real fleet accumulates — a canary batch here, a delayed patch there, a microcode revision that shipped to some racks and not others. A uniform fleet with none of that drift (`examples/fleet-uniform.toml`) is one anonymity set the size of the whole fleet, by construction, and serves only as the tool's control case; Table 5 is the drifted one, more representative of ordinary operation, though both fixtures are synthetic rather than telemetry from a live fleet.

6 Is the Pre-Computation Pool Sound?

Section 5.1 established that, in the desk study's own published figures, pre-computation is what lets the composed path fit its budget at all, not an optional speed-up on top of a design that already worked. That reframing relocates the risk. The question is no longer whether BBS+ is fast enough; it is whether the pool that makes it fast enough is sound. Nobody asked this question before the desk study's arithmetic error made pre-computation load-bearing, because until then it looked like an optimization, and optimizations are not normally where a security review looks.

The two available behaviours. A pool of pre-computed blinding factors sitting in front of a burst of demand has exactly two ways to respond once it is empty: it can make the caller wait for a fresh item, or it can hand back one that has already been used. `occultation`'s `PrecomputationPool` does the first and never the second. On exhaustion, `acquire` returns `None` rather than a used item, and there is no configuration flag or code path under which it returns one. Section 5.2 reports the cost of choosing to stall: under a sustained burst, the delay does not plateau, it grows with the length of the burst, and that growth is a timing side channel with a duration-invariant signature — an observer of an agent's response latency alone can read off how overloaded it is and how long it has been that way.

Why the alternative is not a lesser evil. Reusing one set of blinding scalars across two presentations is not merely a weakening of unlinkability; it discloses the credential outright, and this is provable rather than asserted. A BBS+ presentation's proof of knowledge has the Schnorr-style shape common to sigma protocols: for a hidden value x (the signature's blinding scalar, or an undisclosed attribute) with blinding $\tilde{x}$ and Fiat–Shamir challenge c, the response the verifier sees is

$$\hat{x} = \tilde{x} + c \cdot x \pmod{r}. \tag{1}$$

If two presentations to two different relying parties reuse the same blinding $\tilde{x}$ but necessarily bind to two different challenges $c_1 \neq c_2$ — the challenge is derived from a per-session nonce, so identical challenges would themselves require a coincidence — then subtracting the two responses cancels the blinding entirely:

$$\hat{x}_1 - \hat{x}_2 = (c_1 - c_2)\, x \quad \Longrightarrow \quad x = \frac{\hat{x}_1 - \hat{x}_2}{c_1 - c_2}. \tag{2}$$

This is not a hypothetical weakness. `pool.rs`'s test `reusing_one_set_of_scalars_leaks_the_signature_and_every_hidden_attribute` performs exactly this division on the reused presentation's responses, against a real signature and real hidden attributes, and asserts that the recovered value equals the signature's scalar and every undisclosed attribute in order. The two presentations are also linkable on sight before any arithmetic is done, because they share the same commitment values outright. Reuse is not a smaller version of the unlinkability property failing; it is that property's negation, handed to whoever sees both transcripts, in closed form.

What the type system buys, stated precisely. `PoolItem` is not `Clone` and its `consume` method takes `self` by value, so *accidental* reuse — holding an item, copying it, or re-acquiring it from the pool a second time — is impossible: the fields are private, the type cannot be duplicated, and `acquire` never returns an item to its own queue. This is worth stating exactly because it is easy to overstate. *Deliberate* reuse is not prevented. `consume` places no bound on what its closure returns, and the scalars underneath are an ordinary `Clone`able type with public fields, so `item.consume(|s| s.clone())` yields an owned copy that outlives the item, and the lower-level proving function that accepts pre-chosen scalars is itself public. What the design buys is not a guarantee that reuse cannot happen; it is that any such reuse is one conspicuous, greppable `.clone()` call at the point it happens, rather than an emergent property of a pool running dry under load — which is precisely the failure mode this paper's measurement is about. A code reviewer looking for that call will find it. A load test will not, because there is nothing under load that produces it: exhaustion produces a stall, and only a stall.

What we are not claiming. We have not built or evaluated a third option beyond stall and reuse, and P05's own open question — whether pool refill activity, running as idle-time work inside the protected environment, correlates with agent activity in a way an observable host resource (power draw, cache behaviour, scheduling) could exploit — is untouched by this paper's measurements. `pool`'s simulation also treats refill as a fluid approximation of a single worker rather than a discrete-event queue; the approximation is stated in the code and does not change which of the two behaviours is chosen, but it means the specific latency numbers in Section 5.2 are a model of one refill strategy, not a measurement of every plausible implementation.

7 What We Still Don't Know

Four questions this tool does not answer, in the order the desk study's own open-questions list raises them:

1. **Can a log operator correlate re-randomized commitments by posting time and size, even if the commitments themselves hide everything cryptographically?** The construction's evidence-log layer proposes per-session ephemeral chains with roots posted as re-randomizable Pedersen commitments, specifically so that a log operator cannot reconstruct a global action sequence from the commitments' contents. Nothing in this tool touches that layer: it is not implemented, measured, or modelled here. Cryptographic hiding of a value says nothing about the traffic pattern of when and how often that value is posted, and an operator who can observe posting times and sizes has a side channel that no property of the commitment scheme addresses. This is the specific gap P05 names and the specific thing to prove or refute, and this paper does neither.

2. **What does tamper-evidence mean once the global chain is replaced by independent per-session chains?** A single global hash chain proves that no action was deleted. Independent per-session chains, the construction's own proposal for unlinkability, prove only that no action was deleted *from a session an auditor already knows exists*: an agent that suppresses an entire session leaves no gap for anyone to notice. This may be the more consequential result available from this whole research direction — that unlinkability and completeness-of-record are in tension rather than independently achievable — and this paper's measurements say nothing about it either way, because the evidence-log layer is out of scope here.

3. **Revocation.** Revoking a compromised agent's credential without linking its past actions, and without an accumulator update cost that scales badly with fleet size, is historically the hardest part of deploying anonymous credentials, and it is barely touched by the desk study this paper corrects. `occultation` has no revocation mechanism, measured or modelled.

4. **Is an anonymity set built from TCB configuration diversity even the right set to be measuring?** Section 5.3 answers "how large is the set a real fleet's hardware diversity leaves standing," which is a question worth answering on its own, but it is not the same question as "what is the anonymity set a colluding pair of relying parties should be evaluated against." The right set might be agents within one enterprise, enterprises sharing one credential issuer, or TEE hardware generations across many unrelated fleets, and the answer changes who the scheme protects against whom. Our calculator partitions whatever fleet description it is given; it has no opinion about which fleet is the relevant one, and supplying that opinion is future work, not something this measurement resolves by itself.

Two structural omissions sit above these four and are worth naming rather than leaving implicit. The evidence-log and transport layers of Table 1 are not implemented, measured, or modelled anywhere in this artifact: the first is questions 1 and 2 above, and the second — Oblivious HTTP, relay padding, and everything that stops an unlinkable credential from riding over a directly attributable connection — is named by P05 and untouched by this tool entirely. Every figure this paper reports about cost, exhaustion, or anonymity should be read as a statement about the credential and hardware-attestation layers alone.

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